Model Development Document (MDD)

Project Title:

Time Series Forecasting of Daily Minimum Temperatures Using SARIMAX with Fourier Seasonality

Project Identifier: Capstone Project 2

Author: Bhushan Mandekar

Target Environment: Python 3.11 / Statsmodels / Streamlit Cloud

Dataset Reference: Daily Minimum Temperatures in Melbourne, Australia (1981–1990)

1. Document Control & Metadata

1.1 Software Stack & Dependencies

- **Core Environment:** Python 3.11 (Conda environment: temperature_forecast)
- **Data Processing & Analysis:** pandas, numpy
- **Statistical Modeling & Diagnostics:** statsmodels
- **Visualization:** matplotlib
- **Serving & Interactive Layer:** streamlit
- **Version Control & CI/CD:** Git, GitHub (daily-temperature-forecasting), Streamlit Community Cloud

1.2 Key References

- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. *Time Series Analysis: Forecasting and Control*.
- Hyndman, R. J., & Athanasopoulos, G. *Forecasting: Principles and Practice*.
- Ljung, G. M., & Box, G. E. P. (1978). *On a measure of lack of fit in time series models*. Biometrika, 65(2), 297–303.

2. Executive Summary & Problem Formulation

2.1 Problem Statement & Objectives

Accurately predicting daily minimum temperatures is vital for municipal planning, energy demand estimation, and agricultural logistics. Temperature series exhibit complex seasonality consisting of a long 365.25-day annual cycle combined with short-term, auto-correlated atmospheric shocks. Standard SARIMA specifications become computationally intractable when handling long periods ($m = 365$).

This project formulates an end-to-end forecasting pipeline to model Melbourne's daily minimum temperatures over a 10-year period (1981–1990). The primary model balances deterministic seasonal progression with autoregressive-moving-average stochastic dynamics.

2.2 Core Business & Technical Deliverables

- Cleaned, continuous daily temperature series spanning 1981 to 1990.
- Statistical verification of stationarity, autocorrelation structure, and seasonal decomposition.
- Chronological 3-way partition (Train / Validation / Test) preventing data leakage.
- Multi-model benchmark suite (Naïve, Seasonal Naïve, ARIMA, Holt-Winters, SARIMAX + Fourier).
- Ljung-Box residual diagnostics ensuring white-noise compliance.
- 365-day forward-looking trajectory with 95% prediction intervals.
- Production-ready Streamlit web application deployed via Streamlit Community Cloud.

3. Data Ingestion & Preprocessing

3.1 Dataset Overview

- **Source:** Daily Minimum Temperatures in Melbourne, Australia.
- **Temporal Horizon:** January 1, 1981 – December 31, 1990 (~3,650 daily observations).
- **Raw Schema:** Date (string object), Temperature (numeric string with irregular values).
- **Persistent File:** daily_minimum_temperature_cleaned.csv.

3.2 Data Hygiene & Pipeline Implementation

Inspection revealed corrupted characters (e.g., "?") within raw values. Data cleaning resolved these issues:

1. Removed invalid punctuation and coerced missing readings to numeric datatypes.

2. Converted Date to standardized datetime format, sorted chronologically, and indexed the series at daily frequency (D).

3. Addressed missing observation intervals via linear time-series interpolation to preserve sequence continuity.

Python

```
import pandas as pd

# Data Hygiene Routine
df["Temperature"] = df["Temperature"].astype(str).str.replace("?", "", regex=False)
df["Temperature"] = pd.to_numeric(df["Temperature"], errors="coerce")

# Chronological Indexing & Interpolation
df["Date"] = pd.to_datetime(df["Date"])
df = df.sort_values("Date").drop_duplicates(subset=["Date"])
df = df.set_index("Date").asfreq("D")
df["Temperature"] = df["Temperature"].interpolate(method="time")
```

4. Exploratory Data Analysis & Time-Series Diagnostics

4.1 Global Distribution & Summary Statistics

The overall series fluctuates between 0.0°C and 26.3°C, exhibiting a roughly Gaussian profile centered around an empirical median of 11.0°C.

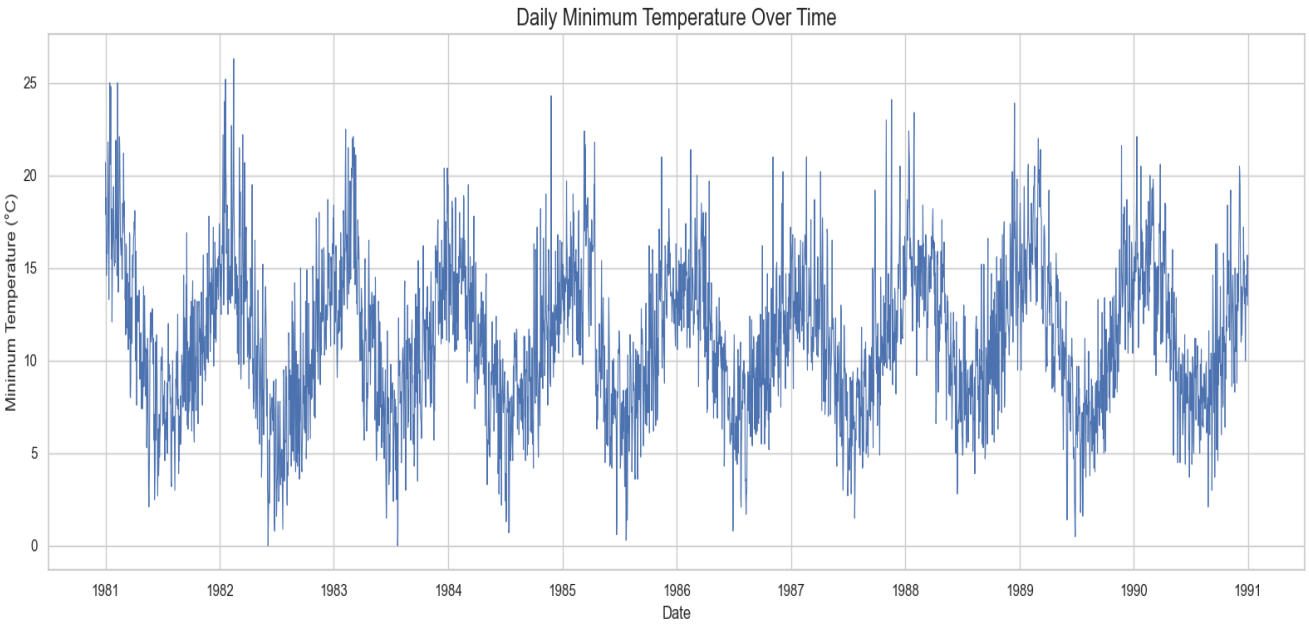


Figure 1: Full 10-year historical daily minimum temperature record (1981–1990), highlighting stable bounded cycles.

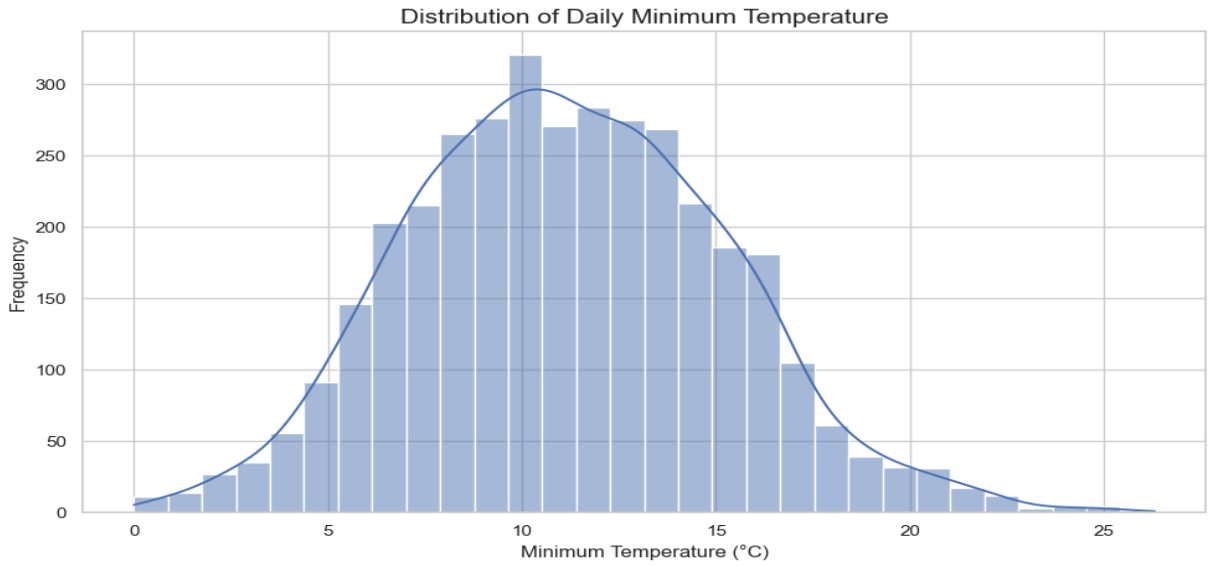


Figure 2: Empirical distribution histogram and KDE density plot showing near-normal distribution of daily temperatures.

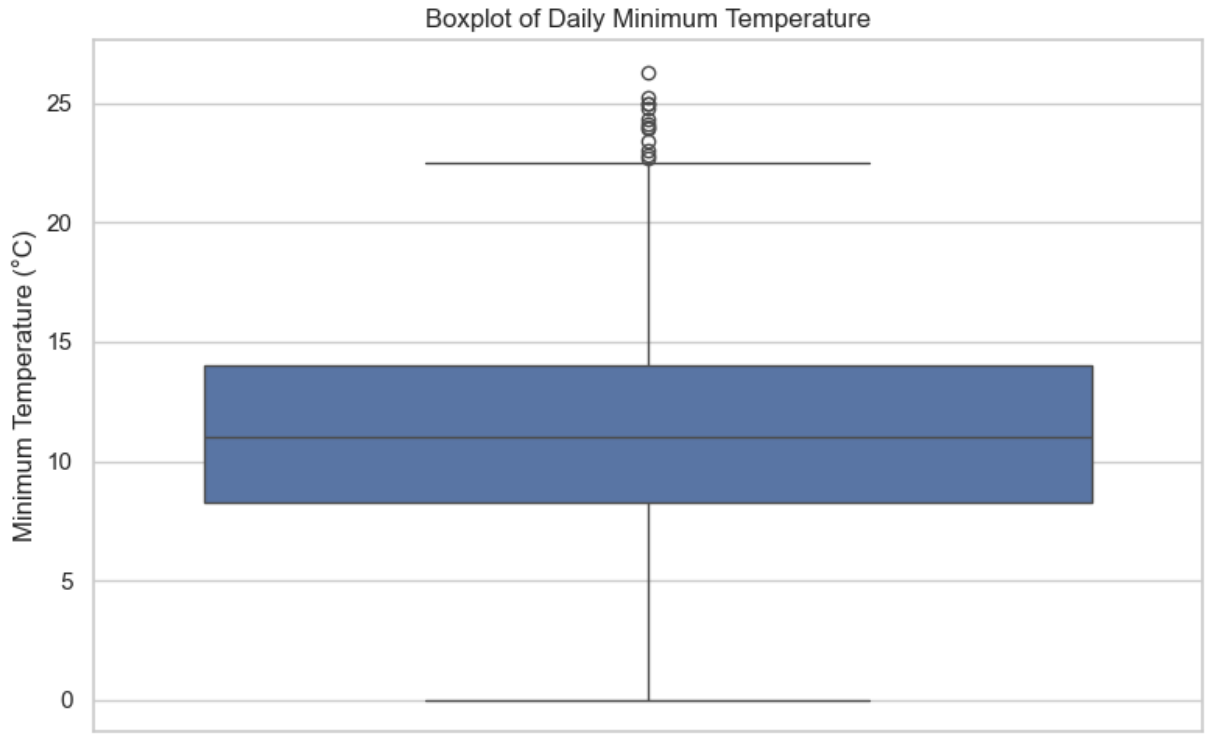


Figure 3: Global boxplot identifying upper boundary values (>23°C). Observations are retained as valid meteorological anomalies rather than measurement errors.

4.2 Seasonality & Rolling Temporal Variance

Monthly and seasonal groupings confirm strong annual seasonality: temperatures peak in Southern Hemisphere summer (January–February) and reach annual lows in winter (June–July). Rolling statistics show cyclical mean shifts while standard deviations remain stationary between 1.5°C and 4.0°C.

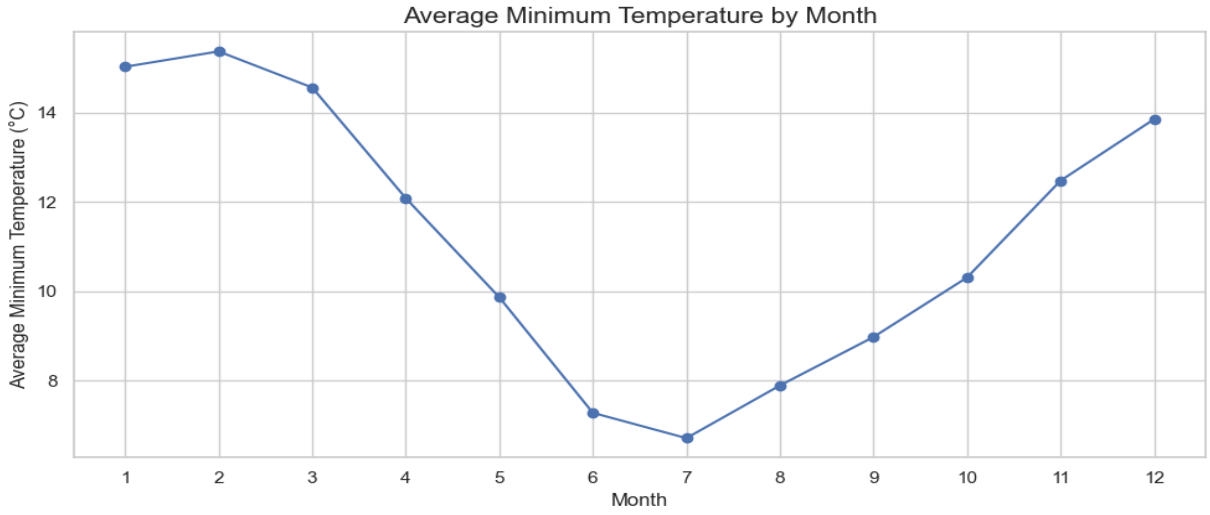


Figure 4: Monthly mean trajectory demonstrating sinusoidal variation peaking in February (~15.5°C) and bottoming in July (~6.8°C)

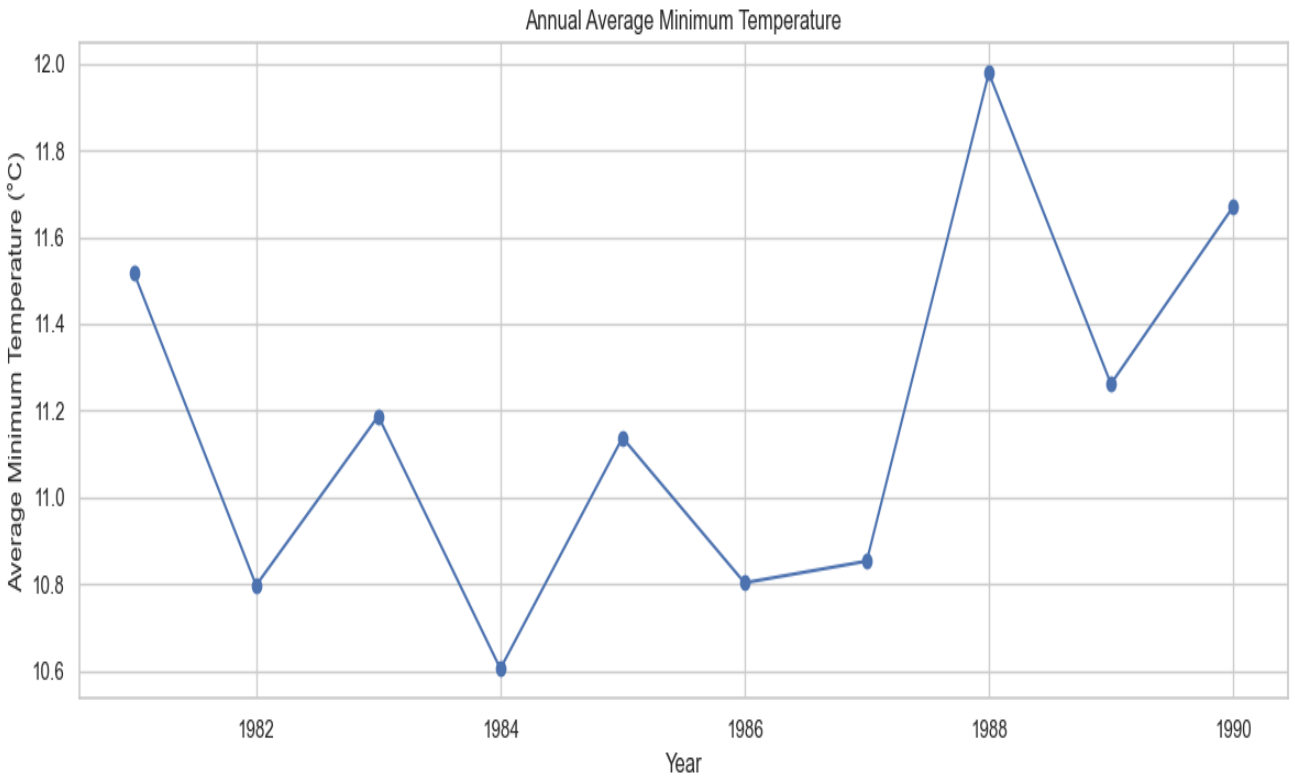


Figure 5: Annual mean temperatures across 1981–1990 highlighting year-over-year baseline stability (10.6°C to 12.0°C).

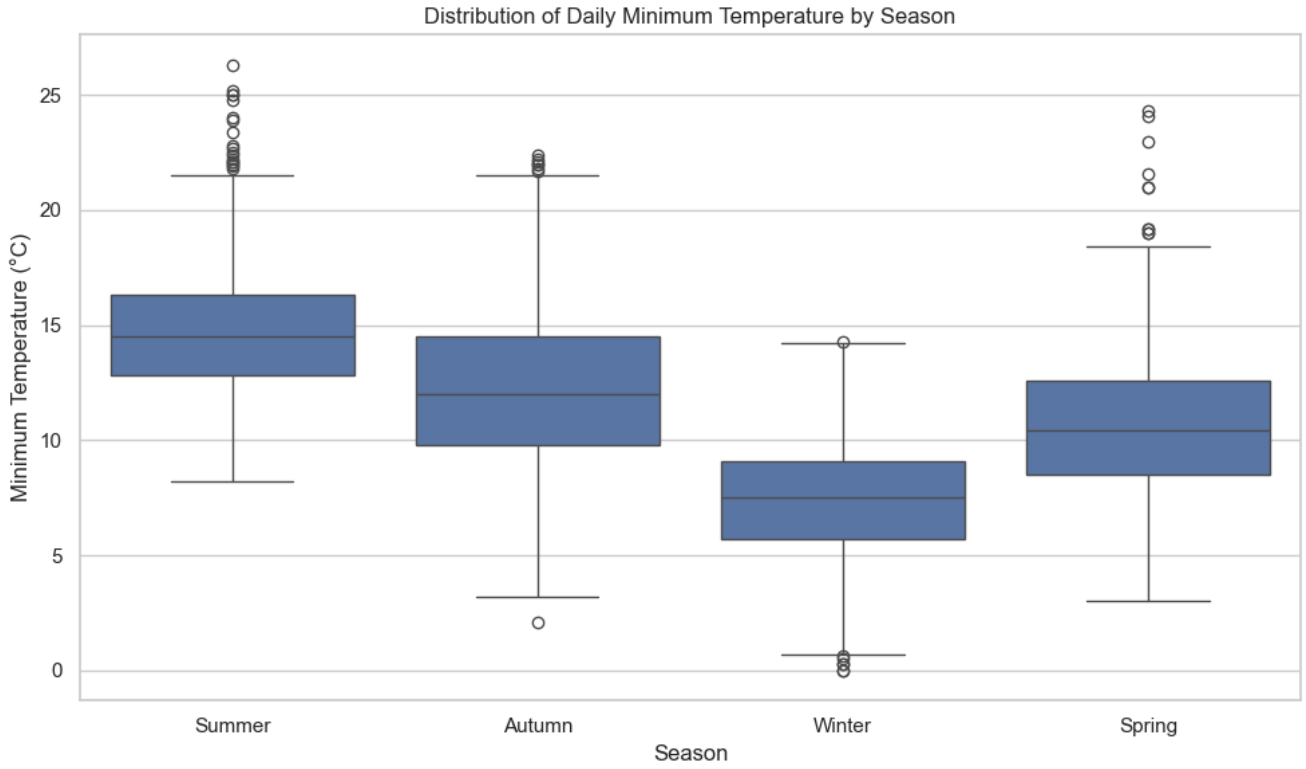


Figure 6: Seasonal temperature boxplots illustrating distinct shifts across Summer, Autumn, Winter, and Spring.

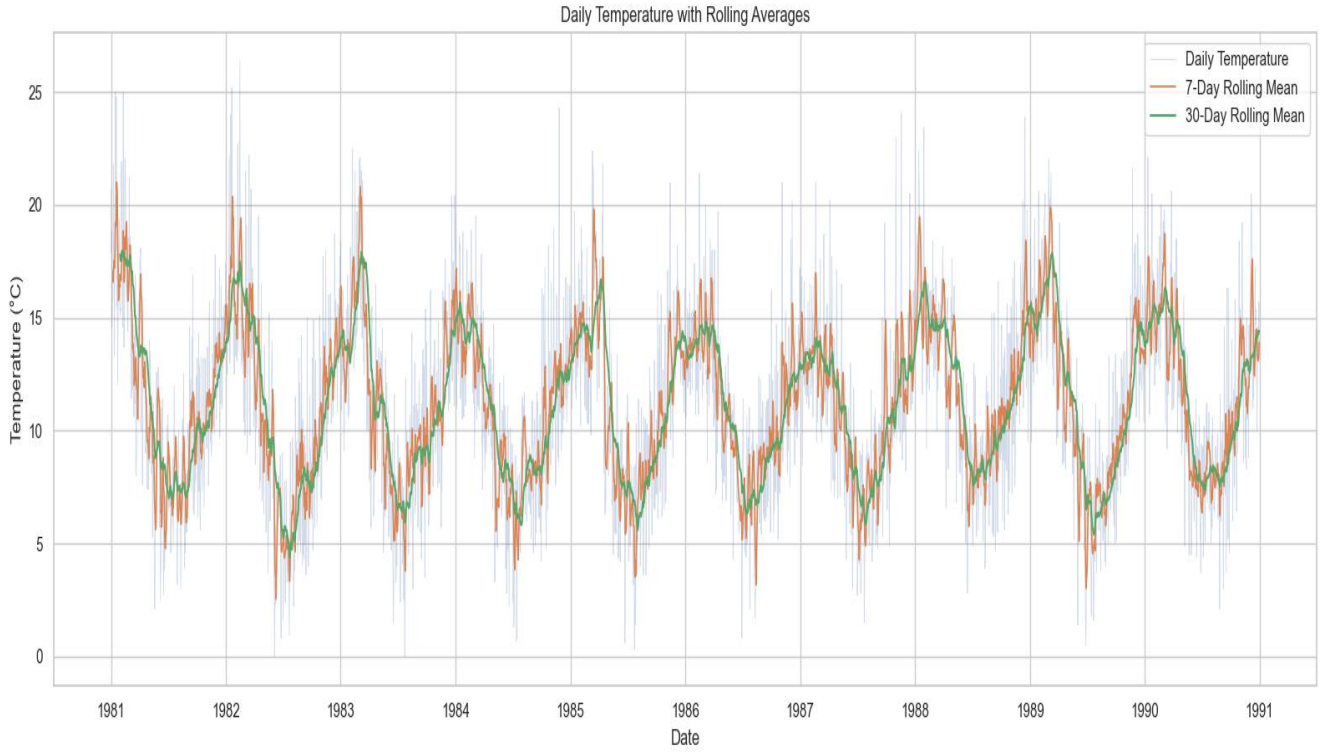


Figure 7: 7-day and 30-day moving averages demonstrating short-term smoothing overlaid against the underlying daily series.

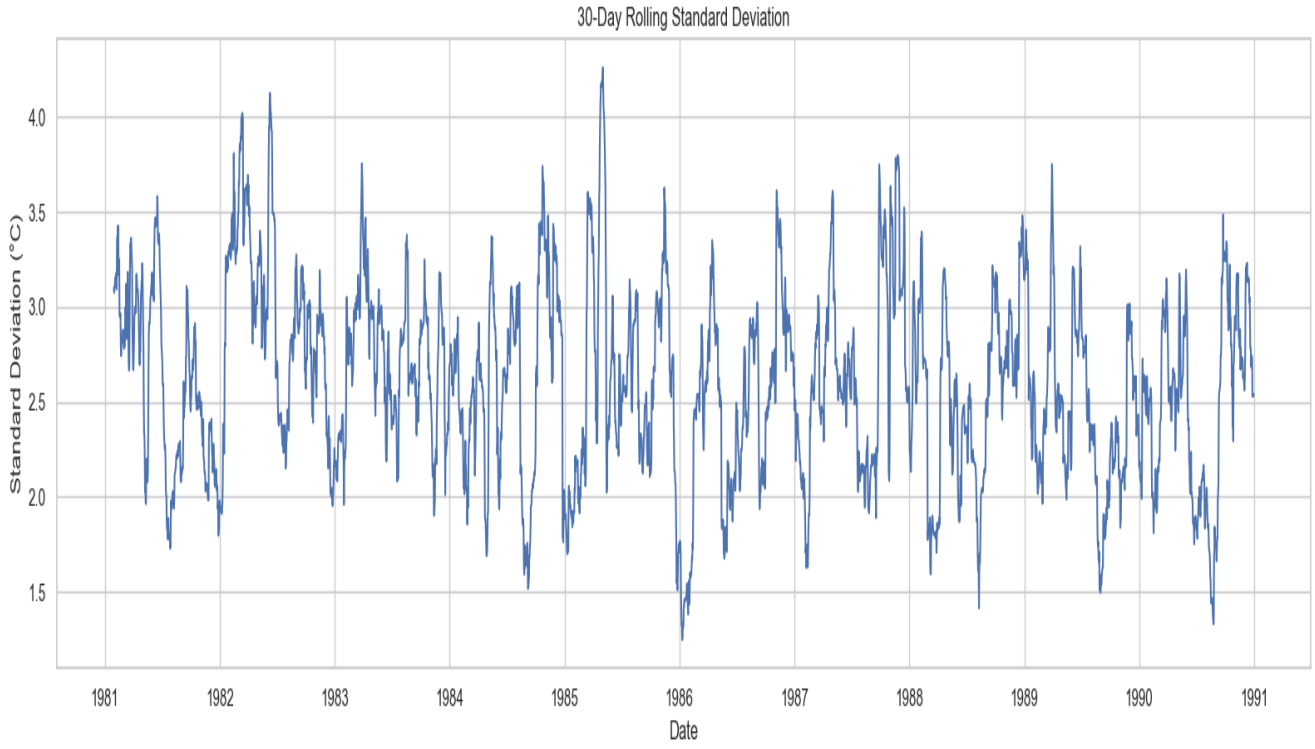


Figure 8: Rolling standard deviation capturing temporal fluctuations in volatility across seasons.

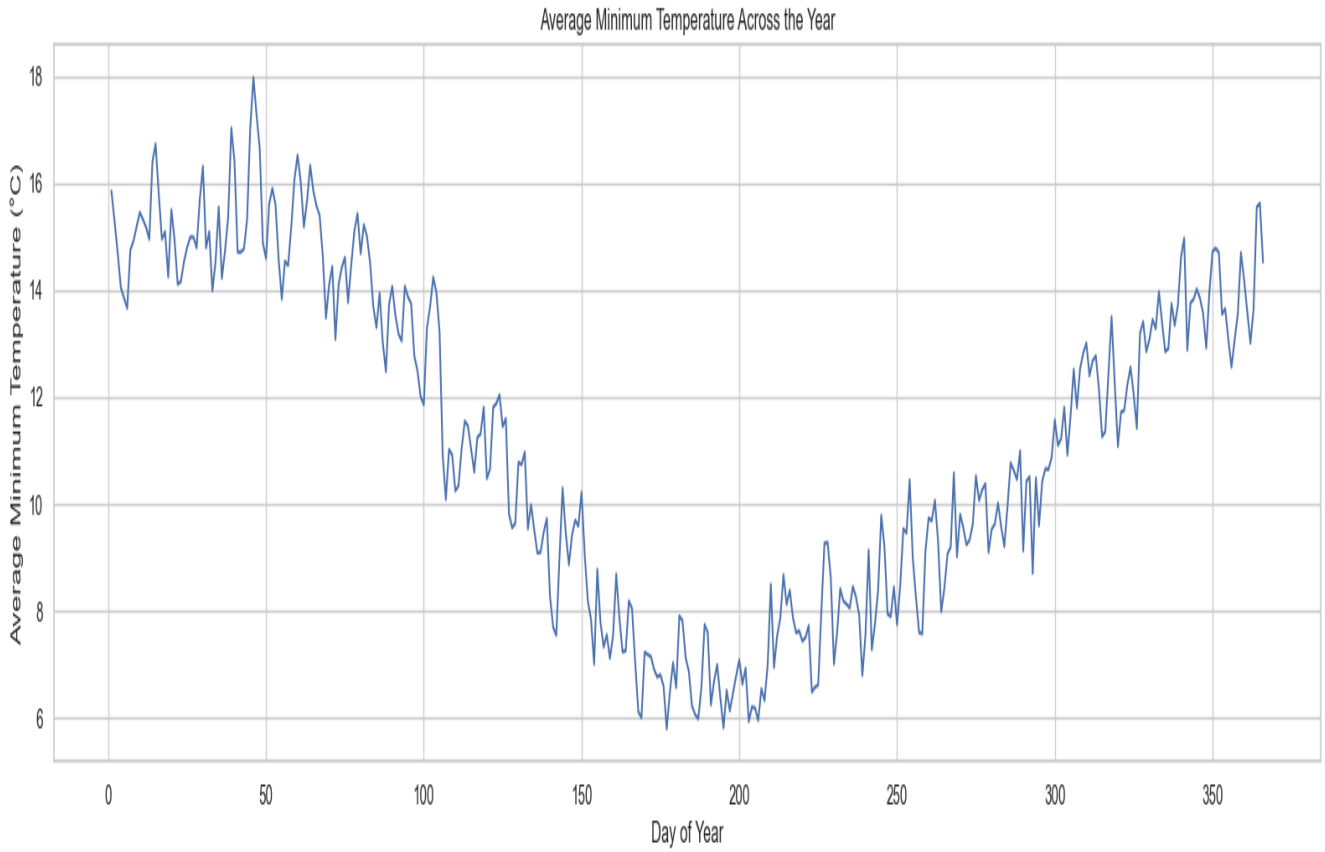


Figure 9: Day-of-year aggregated profile across all 365 calendar days confirming a continuous sinusoidal pattern.

4.3 Stationarity & Autocorrelation Structure

- **Autocorrelation Function (ACF):** Shows slow decay across extended lags (1, 2, 3, 7, 14, 30, 90, 180, 365 days), confirming strong seasonal persistence.

- **Partial Autocorrelation Function (PACF):** Shows a sharp spike at lag 1 followed by secondary activity up to lag 3 before falling within Bartlett confidence bands.

- **Stationarity (ADF Test) & Differencing Strategy:** While first-differencing achieves strict stationarity, explicitly modeling seasonal dynamics via Fourier exogenous regressors allows the target series to be modeled directly at level ($d = 0$), preserving interpretable temperature scales.

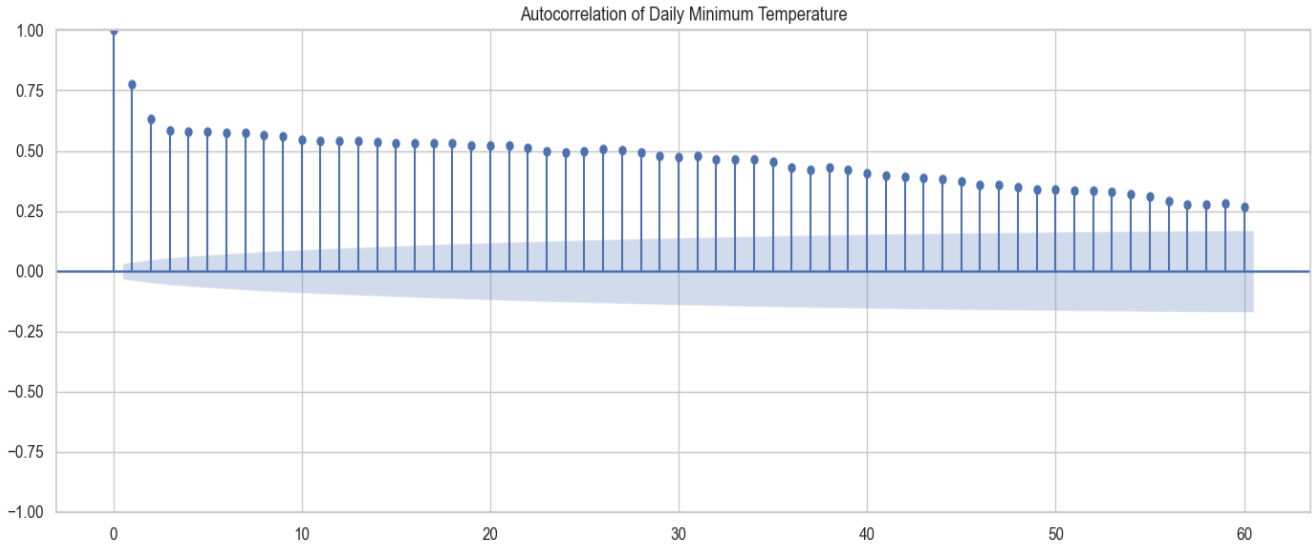


Figure 10: Sample ACF demonstrating high persistence up to lag 60 due to unmodeled annual seasonality.

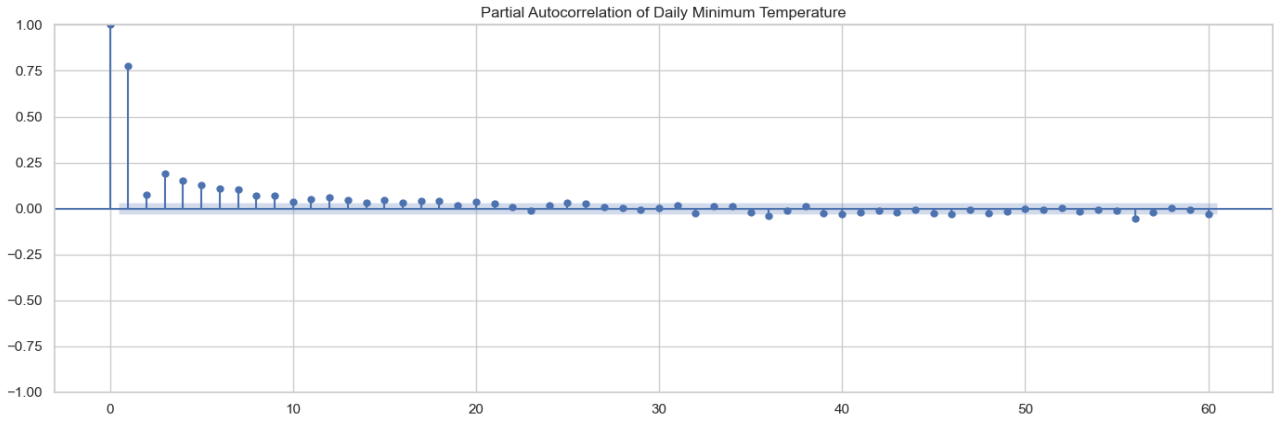


Figure 11: PACF showing dominant autoregressive dependence concentrated in the initial 1 to 3 lags.

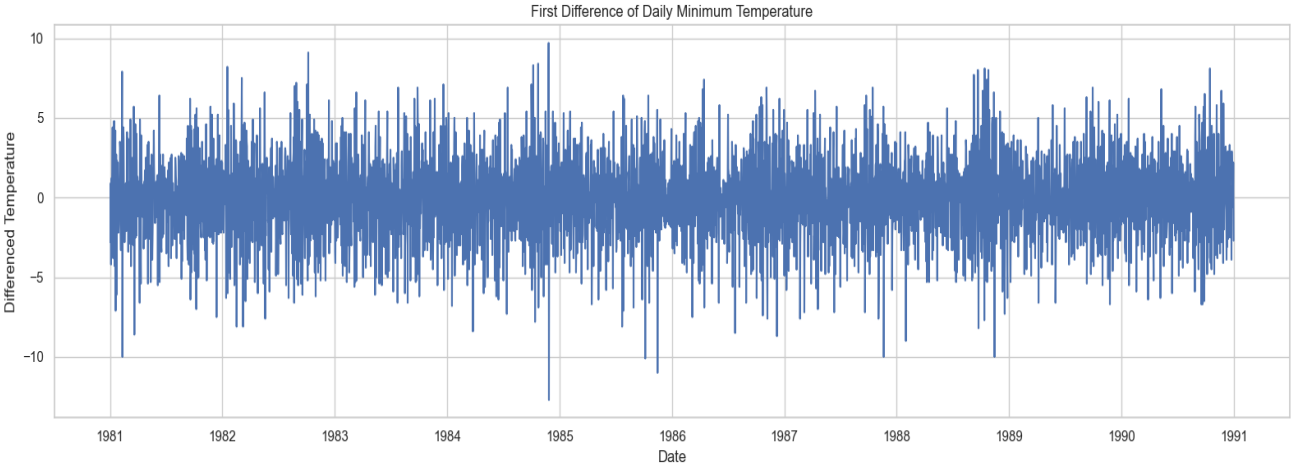


Figure 12: Stationary first-differenced series (Δy_t) with stabilized mean centering around zero.

5. Mathematical Formulation & Feature Engineering

5.1 Fourier Seasonality Decomposition

Direct SARIMA seasonal orders with period $m = 365$ require hundreds of seasonal autoregressive/moving-average coefficients, leading to computational instability. Instead, annual periodicity is modeled via a finite sum of $K = 5$ Fourier harmonic sinusoids:

$$S_t = \sum_{k=1}^{K} \left[\alpha_k \sin\left(\frac{2\pi k t}{365.25}\right) + \beta_k \cos\left(\frac{2\pi k t}{365.25}\right) \right]$$

Where:

- $t \in [1,2,\dots,N]$ represents the cumulative daily index.
- The base period is set to $P = 365.25$ to account for leap-year drift.
- $K = 5$ generates 10 exogenous columns (sin_1, cos_1, ..., sin_5, cos_5) that provide smooth approximation of the annual curve.

5.2 SARIMAX Formulation

The final process models daily temperature y_t as a regression with SARIMA errors:

$$y_t = X_t \gamma + \eta_t$$

$$\phi(B)(1-B)^d \eta_t = \theta(B)\epsilon_t, \quad \epsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

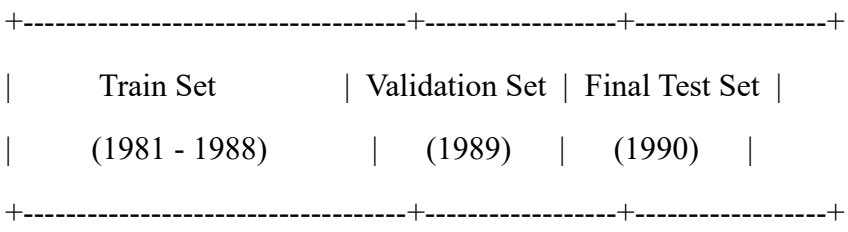
Where:

- X_t denotes the row vector of 10 Fourier harmonic regressors.
- γ is the vector of estimated regression coefficients.
- $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3$ is the AR(3) polynomial operator.
- $\theta(B) = 1 + \theta_1 B$ is the MA(1) polynomial operator.
- Integration order $d = 0$.

6. Model Training & Validation Framework

6.1 Chronological Data Split Scheme

To prevent lookahead bias, the series was split into three contiguous chronological windows:



- **Training Window (1981–1988):** 8 years of daily data used for parameter estimation.
- **Validation Window (1989):** 1 full seasonal cycle (365 days) used for model architecture selection, grid search tuning, and residual diagnostics.
- **Held-out Test Window (1990):** 1 full seasonal cycle (365 days) retained untouched until final testing.

6.2 Evaluation Metrics

Models were scored on predictive precision:

- **Mean Absolute Error (MAE):** Reflects average error magnitude in degrees Celsius.
- **Root Mean Squared Error (RMSE):** Penalizes extreme deviations more heavily.
- *Note:* Mean Absolute Percentage Error (MAPE) was omitted because near-zero winter temperatures inflate percentage errors.

$$\text{MAE} = \frac{1}{H}\sum_{h=1}^{H} |\, y_{T+h} - \hat{y}_{T+h}|, \quad \text{RMSE} = \sqrt{\frac{1}{H}\sum_{h=1}^{H} (y_{T+h} - \hat{y}_{T+h})^2}$$

6.3 Baseline & Candidate Model Evaluations

Model 1: Naïve Baseline

Assumes tomorrow's minimum temperature equals today's observed value ($\hat{y}_{T+h} = y_T$). Fails across long horizons as forecasts flatten into a constant horizontal line.

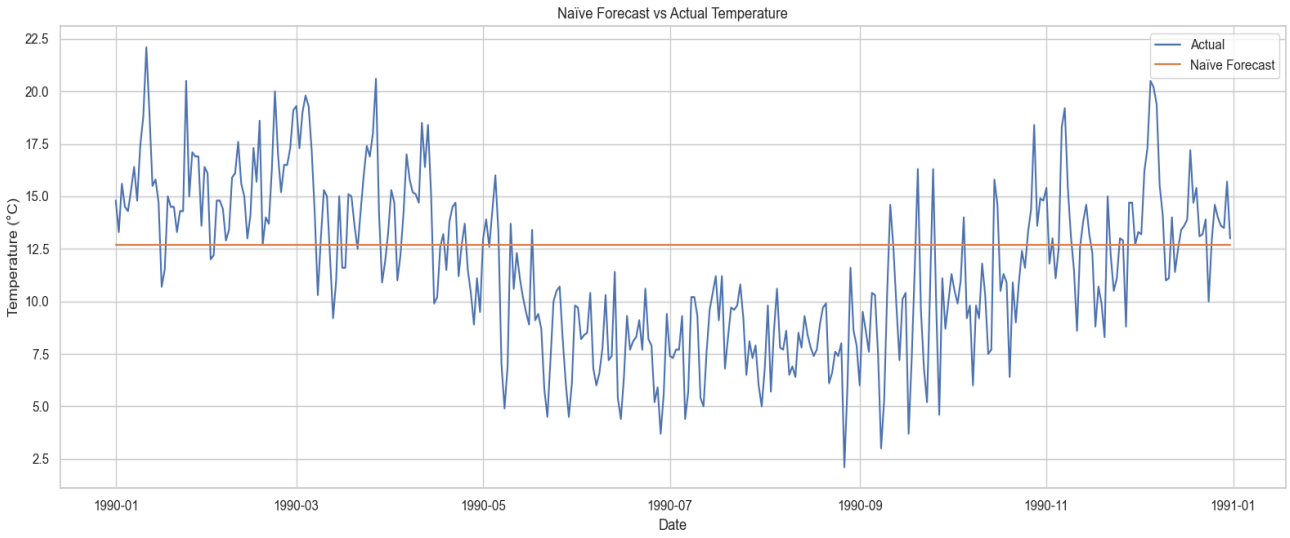


Figure 13: Naïve baseline forecast on the evaluation set, demonstrating rapid decay to a flat projection.

Model 2: Seasonal Naïve Baseline

Projects temperature from the corresponding day of the preceding year ($\hat{y}_{T+h} = y_{T+h-365}$). Provides a benchmark by naturally mirroring seasonality, but repeats prior anomalous noise:

- **Validation Performance:** MAE = 2.87°C, RMSE = 3.65°C.

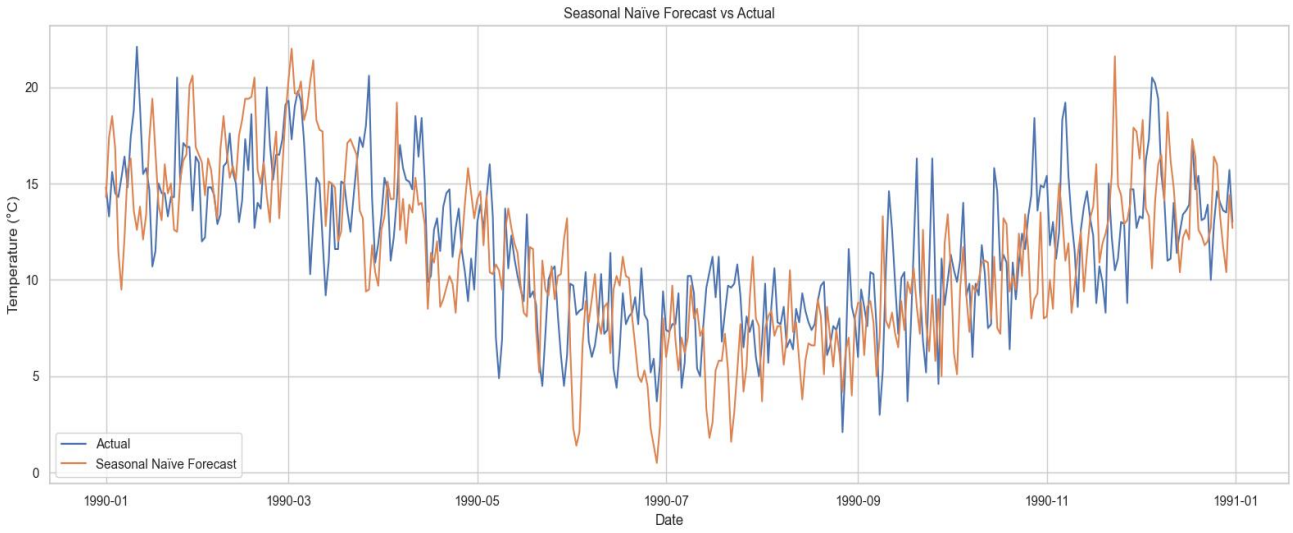


Figure 14: Seasonal Naïve forecast tracking the annual cycle, but replicating single-day anomalies.

Model 3: Standard Non-Seasonal ARIMA

Grid-searched across $p \in [0,3], d \in [0,1], q \in [0,3]$. Top candidate: **ARIMA(1,0,2)**. Because standard ARIMA lacks seasonal mechanisms for period 365, long-range forecasts revert to the empirical series mean:

- **Validation Performance:** MAE $= 3.20°C$, RMSE $= 3.86°C$.

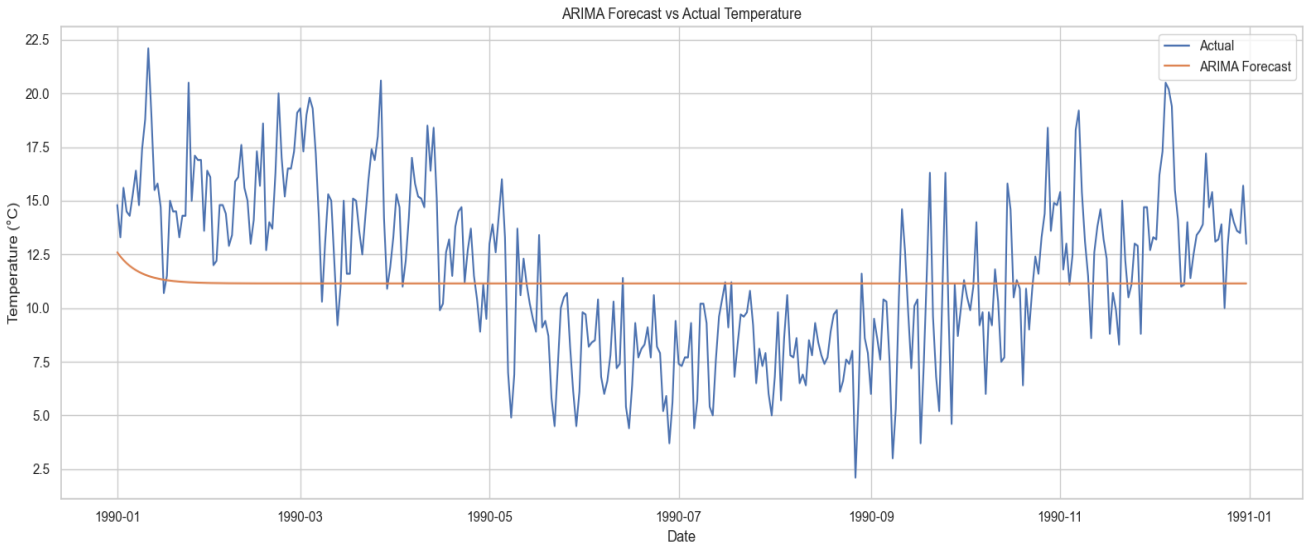


Figure 15: ARIMA(1,0,2) projection rapidly decaying toward the unconditional sample mean.

Model 4: Holt-Winters Exponential Smoothing

Additive Holt-Winters model fitted across seasonal cycles. Underperformed baseline metrics across daily intervals:

- **Validation Performance:** MAE $= 4.1575°C$, RMSE $= 4.8514°C$.

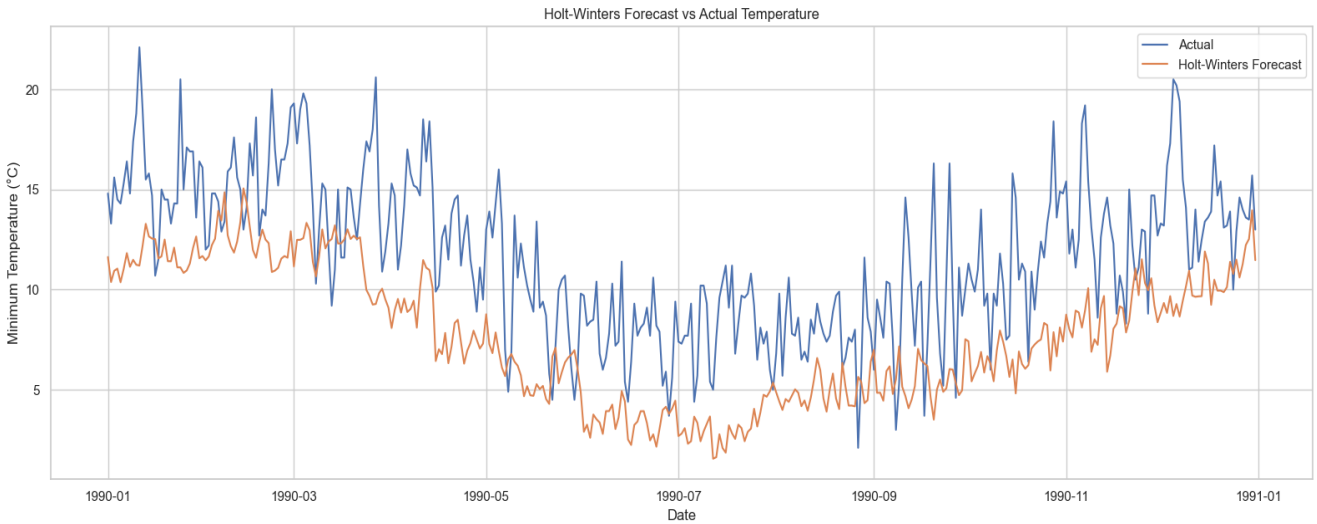


Figure 16: Holt-Winters exponential smoothing forecast exhibiting seasonal bias and elevated deviation.

7. Model Selection & Statistical Validation

7.1 Validation Comparison (1989 Set)

Initial candidate screening on the 1989 validation set:

Model Architecture	MAE (°C)	RMSE (°C)	Diagnostic Summary
SARIMAX(1,0,2) + Fourier	2.0432	2.6482	Lowest initial RMSE, but failed residual autocorrelation tests.
Seasonal Naïve	2.8674	3.6508	Strong heuristic baseline.
ARIMA(1,0,2)	3.1980	3.8622	Reverts to historical mean.
Naïve Baseline	3.3236	3.9916	Flat forecast.
Holt-Winters	4.1575	4.8514	High systematic variance.

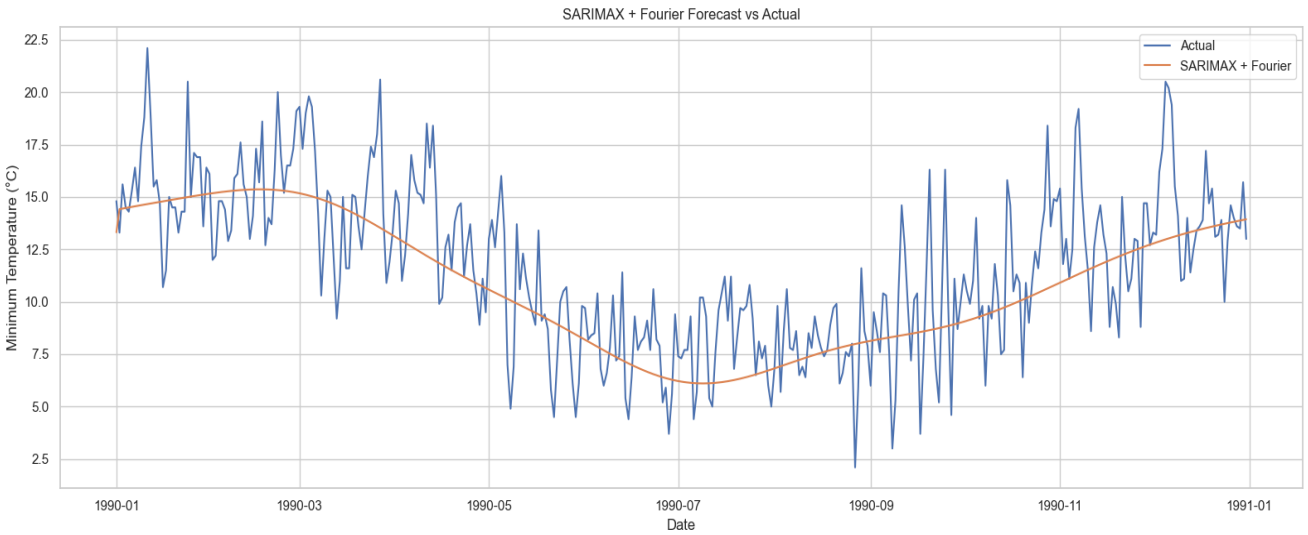


Figure 17: SARIMAX model incorporating Fourier harmonics, capturing annual cyclical variation.

7.2 Residual Diagnostics & Ljung-Box Analysis

While **SARIMAX(1,0,2) + Fourier** yielded marginally lower raw validation RMSE, residual diagnostics revealed significant remaining autocorrelation. Autocorrelated residuals violate the white-noise assumption ($\epsilon_t \sim WN(0, \sigma^2)$) and lead to unreliable prediction intervals.

Higher-order candidate models were evaluated using the Ljung-Box portmanteau test across diagnostic lags 7, 14, and 30 (H_0: residuals are independently distributed):

Model Specification	**Lag 7 (p-value)**	**Lag 14 (p-value)**	**Lag 30 (p-value)**	**White Noise Status**
SARIMAX(1,0,2) + Fourier	< 0.001	< 0.001	< 0.001	**Rejected** (correlated errors)
SARIMAX(3,0,3) + Fourier	0.425	0.735	0.918	Passed (white noise)
SARIMAX(2,0,2) + Fourier	0.435	0.737	0.913	Passed (white noise)
SARIMAX(3,0,1) + Fourier	**0.663**	**0.871**	**0.954**	**Passed (optimal white noise & parsimony)**

SARIMAX(3,0,1) with $K = 5$ Fourier harmonics was selected as the final production model. It showed no residual autocorrelation across all tested lags, maintaining optimal model parsimony and lower BIC.

8. Final Test Evaluation & Benchmarking (1990 Held-Out Data)

The selected model was evaluated against the unseen 1990 test dataset.

8.1 Final Out-of-Sample Performance Table

Rank	Model Architecture	MAE (°C)	RMSE (°C)	RMSE Improvement vs. Baseline
1	**SARIMAX(3,0,1) + Fourier (Selected)**	**2.0238**	**2.6263**	**+28.06% vs Seasonal Naïve**
2	Seasonal Naïve	2.8674	3.6508	Baseline Benchmark
3	ARIMA(1,0,2)	3.1982	3.8625	-5.79%
4	Naïve Baseline	3.3236	3.9916	-9.33%
5	Holt-Winters Exponential Smoothing	3.5425	4.2043	-15.16%

Rounded final test metrics: MAE ≈ 2.02°C and RMSE ≈ 2.63°C. When compared to the Seasonal Naïve baseline, this reflects a **29.42% reduction in MAE** and a **28.06% reduction in RMSE**.

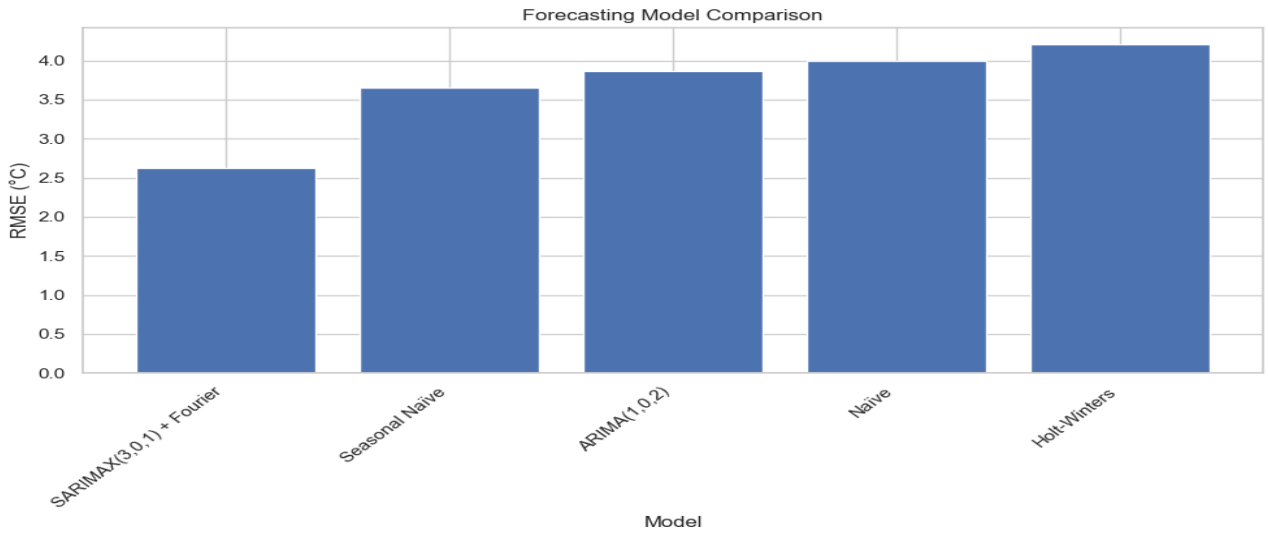


Figure 18: Comparative RMSE bar chart illustrating the error reduction achieved by SARIMAX(3,0,1) + Fourier.

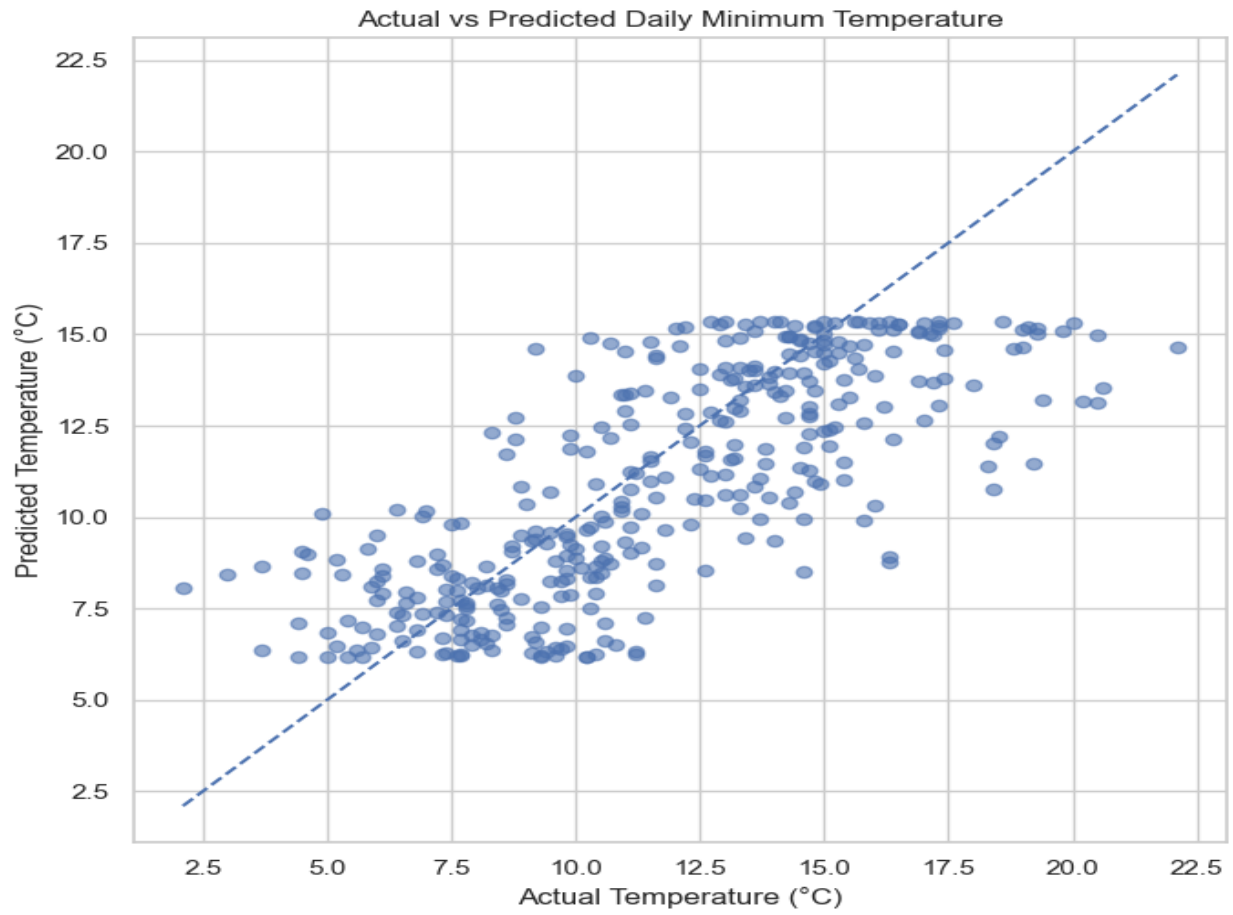


Figure 19: Parity scatter plot displaying predicted versus actual temperatures clustering along the line of equality ($y = x$).

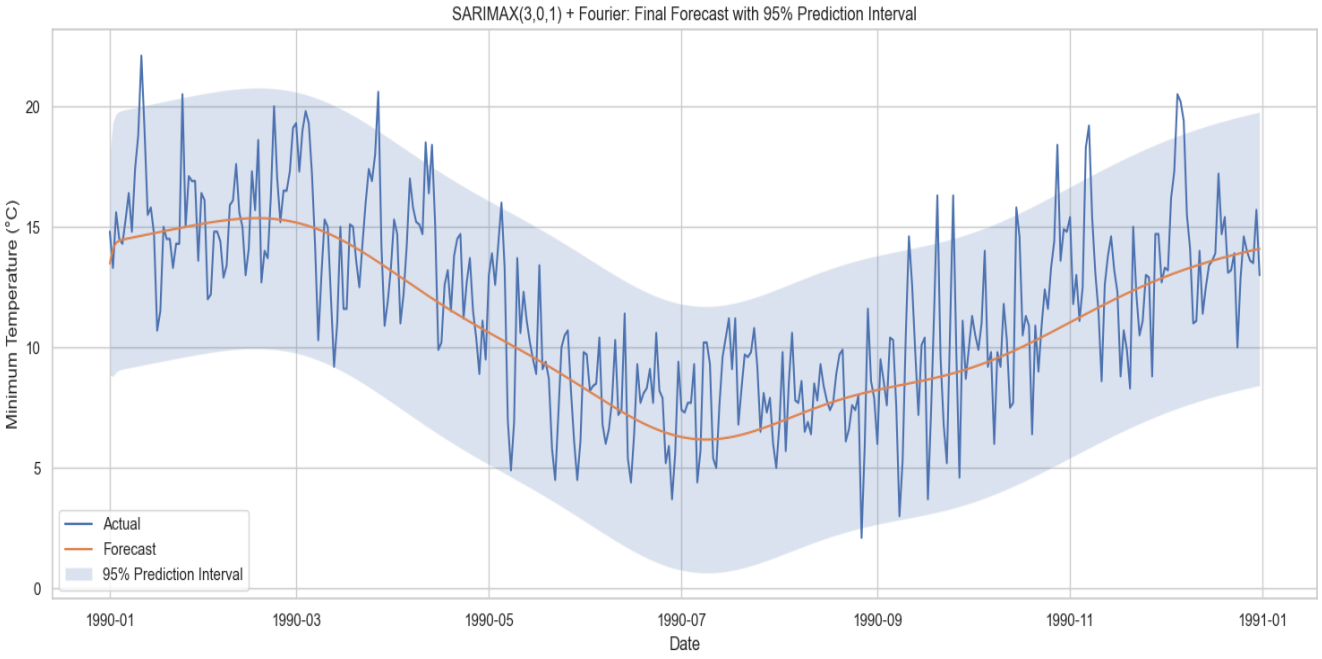


Figure 20: 1990 out-of-sample forecast with 95% prediction intervals, tracking the seasonal cycle.

8.2 Error Analysis & Extreme Event Inspection

Analysis of the largest forecast errors identified specific days with anomalies exceeding ± 7°C:

- **1990-01-11 & 1990-11-07:** Sudden, unseasonal summer heatwaves.
- **1990-08-27, 1990-09-20 & 1990-10-28:** Cold fronts producing unexpected winter and spring temperature drops.

Because this univariate formulation relies strictly on historical temperature data without real-time pressure, wind direction, or cloud cover covariates, it models the expected seasonal path rather than sudden atmospheric shifts.

9. Production Model Refitting & Forward Forecast

9.1 Model Refitting & Serialization

Following validation, the SARIMAX(3,0,1) + 5 Fourier harmonics architecture was refit on the complete 10-year dataset (1981–1990) to maximize parameter precision.

- **Output Artifact:** models/sarimax_fourier_model.pkl
- **Output Datasets:**
    - SARIMAX_Fourier_365_day_forecast.csv
    - forecast_model_comparison.csv
    - monthly_temperature_forecast.csv

9.2 365-Day Forward Forecast Summary

The model generated an out-of-sample forecast for the next 365 calendar days:

- **Expected Annual Mean:** ≈ 11.49°C (Median: ≈ 11.40°C)
- **Expected Annual Minimum:** ≈ 6.88°C (Winter low)
- **Expected Annual Maximum:** ≈ 15.87°C (Summer peak)

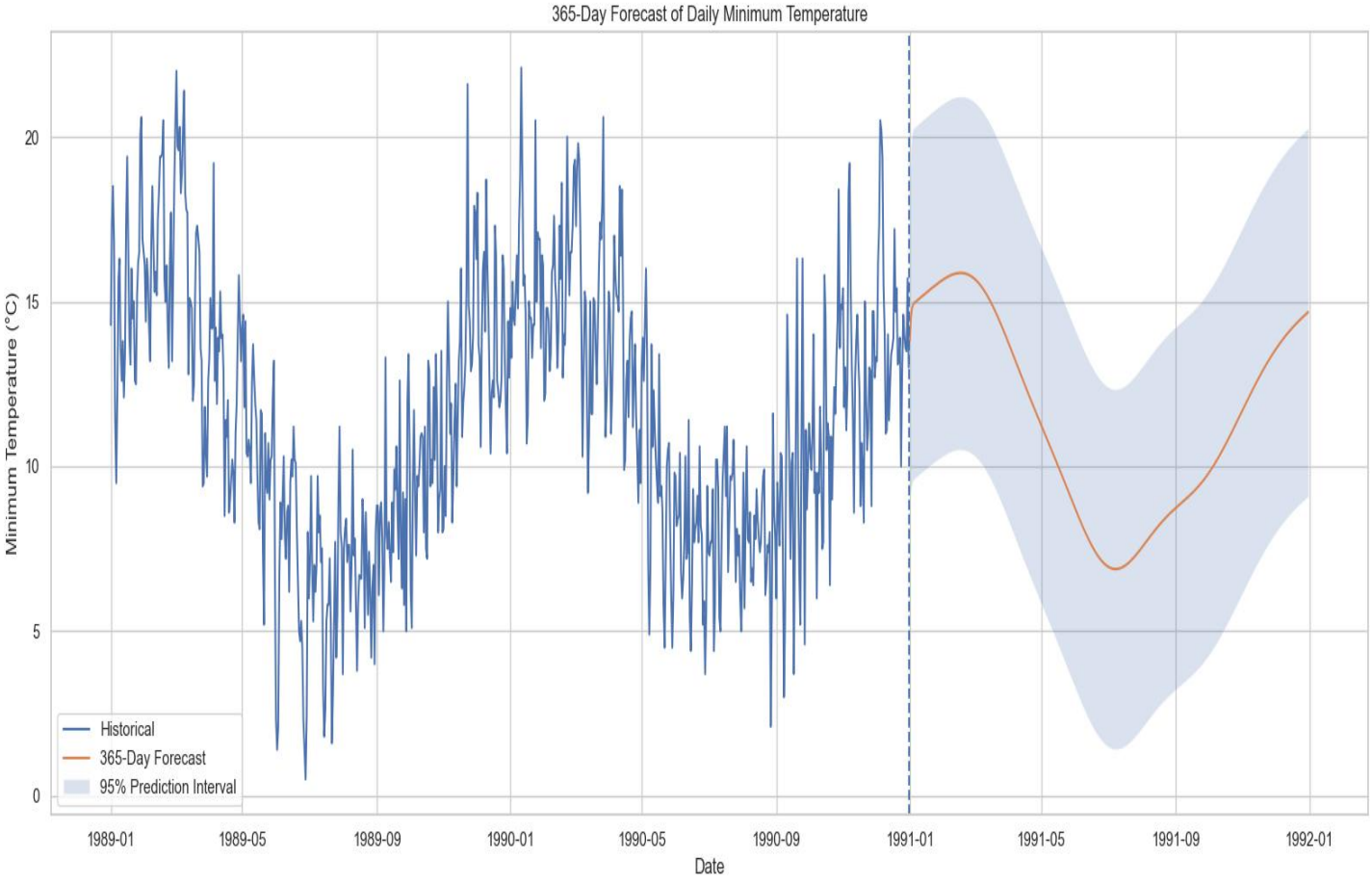
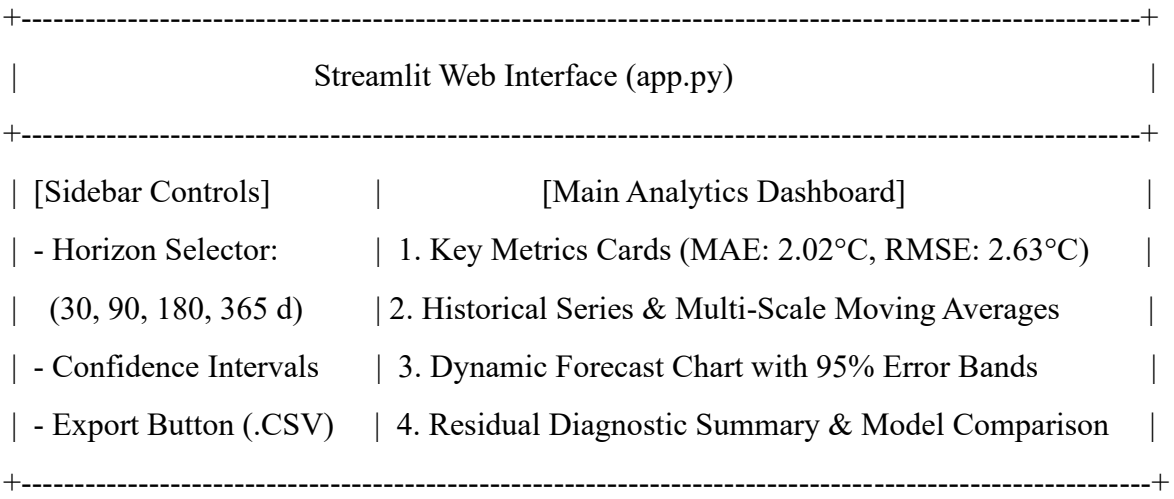


Figure 21: Full 365-day forward forecast showing seasonal progression and widening 95% confidence intervals.

10. Deployment Architecture & User Interface

10.1 Streamlit Interactive Dashboard

An interactive Streamlit application (app.py) was developed to serve the model without requiring local technical tooling.



10.2 Local Testing & Path Resolution

Initial local testing encountered an absolute path resolution failure. This was resolved by using directory-relative path resolutions:

Python

```
from pathlib import Path

# Robust Path Resolution
BASE_DIR = Path(__file__).resolve().parent
DATA_FILE = BASE_DIR / "daily_minimum_temperature_cleaned.csv"
MODEL_FILE = BASE_DIR / "models" / "sarimax_fourier_model.pkl"
```

10.3 Cloud Production Deployment

- **GitHub Repository:** daily-temperature-forecasting

- **Hosting Platform:** Streamlit Community Cloud

- **Target Configuration:** Branch: main, Execution Entry: app.py, Dependency Manifest: requirements.txt

11. Limitations & Risk Mitigation

- **Univariate Boundary:** The model relies solely on historical daily minimum temperature, omitting covariates such as atmospheric pressure, humidity, and cloud cover.

    - *Mitigation:* Integrate exogenous meteorological variables (multivariate SARIMAX) when secondary sensor data is available.

- **Extreme Weather Outliers:** The Fourier regression models smooth seasonal paths and cannot anticipate sudden synoptic weather events.

    - *Mitigation:* Pair long-horizon Fourier projections with short-horizon (1–3 day) numerical weather prediction (NWP) models.

- **Long-Horizon Forecast Uncertainty:** Prediction intervals widen noticeably across the 365-day forecast horizon.

    - *Mitigation:* Use rolling forecasts that refit or update states as new daily observations arrive.

- **Historical Distribution Shifts:** Historical data spans 1981–1990; long-term climate trends may shift baseline distributions over extended time horizons.

    Mitigation: Implement data drift monitoring that flags changes in annual mean minimums.

12. Conclusion & Operational Sign-Off

The **SARIMAX(3,0,1) model with 5 Fourier harmonics** successfully captures both annual seasonality and short-term autocorrelation. On the completely held-out 1990 test set, the model achieved an **MAE of 2.02°C** and an **RMSE of 2.63°C**, outperforming the Seasonal Naïve baseline by **28.06%** while passing all Ljung-Box residual white-noise tests. The serialized model and interactive Streamlit web dashboard are documented and ready for production use.